Aloïs Duston

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alduston.github.io | GitHub | Google Scholar | LinkedIn

Ph.D. researcher developing diffusion and latent generative models, variance-reduced score estimators, and samplers for Bayesian inverse problems. Combines mathematical analysis with PyTorch implementation and experiments in image generation and scientific machine learning.

EDUCATION

The University of Texas at Austin Sep 2024 - Present
Ph.D., Computational Science, Engineering, and Mathematics
Oden Institute | Advisor: Tan Bui-Thanh | CSEM Graduate Fellowship

University of Washington Sep 2022 - Dec 2023
M.S., Applied Mathematics | GPA: 3.98/4.00
Advisors: Bamdad Hosseini and J. Nathan Kutz
Thesis: *Kernel Methods for Conditional Generative Modeling*

University of Virginia Aug 2016 - May 2020
B.A., Mathematics; minor in Biology | Major GPA: 3.93/4.00

PUBLICATIONS AND PREPRINTS

Laplace–Fisher Gate Identities for Optimal Matrix-Gated Blended Score Estimation
A. Duston and T. Bui-Thanh. Preprint, 2026. arXiv:2606.25169.

Variance-Reduced Diffusion Sampling via Target Score Identity
A. Duston and T. Bui-Thanh. Preprint, 2026. arXiv:2601.01594.
Submitted to *SIAM Journal on Scientific Computing*.

Quantifying Single and Dual Channel Live Imaging Data: Kymograph Analysis of Organelle Motility in Neurons
L. Digilio, L. P. McMahon, **A. Duston**, C. C. Yap, and B. Winckler.
Bio-protocol 13(10), e4675, 2023. doi:10.21769/BioProtoc.4675.

RESEARCH EXPERIENCE

Research Assistant, PHO-ICES Group Dec 2023 - Present
The University of Texas at Austin | Advisor: Tan Bui-Thanh

End-to-end latent diffusion with CSEM *Technical report; experiments ongoing*

- Developed Component Score Expectation Matching: analytic supervision for jointly training a variational autoencoder and latent diffusion prior, removing auxiliary denoising noise from the target.
- Implemented flat and hierarchical models in PyTorch; evaluated generation and reconstruction on Fashion-MNIST and CIFAR-10 using FID and capacity-matched comparisons.

Gated Action Dissipation for Bayesian inverse problems *Manuscript in preparation*

- Developed a training-free iterative diffusion sampler that rebuilds scores from target derivatives and an unweighted sample bank, without evaluating the transported density.
- Established one-round Gaussian exactness and local KL decrease conditions; benchmarked Darcy and Navier–Stokes posteriors against MALA at matched PDE and derivative budgets.

Variance-reduced score estimation *Two first-author preprints*

- Derived variance-optimal scalar and matrix gates for blending Tweedie and target-score estimators; established finite-reference consistency and stability bounds.
- Implemented reverse-SDE and probability-flow ODE methods for sampling, normalized posterior-density evaluation, and model-evidence diagnostics in inverse problems.

RESEARCH EXPERIENCE CONTINUED

Research Assistant, Hosseini Research Group Apr 2023 - Aug 2024
University of Washington | Advisor: Bamdad Hosseini

- Developed triangular kernel transport maps for conditional generative modeling and likelihood-free Bayesian inference; evaluated performance against neural generative baselines.
- Derived maximum mean discrepancy (MMD) bounds for posterior approximations induced by generative priors.

Research Assistant, Kutz Research Group Jan 2023 - Aug 2024
University of Washington | Advisor: J. Nathan Kutz

- Developed latent ensemble regularization for deep autoencoders and autoencoder SINDy, improving robustness of physical-law discovery from sparse, noisy data and video.

Research Assistant, Winckler Lab Dec 2020 - May 2021
University of Virginia School of Medicine

- Built KymoMerge, a Python utility for dual-channel neuronal-image processing and statistical analysis; coauthored the published analysis protocol.

INDUSTRY AND SCIENTIFIC COMPUTING EXPERIENCE

Computational Scientist, Research Computing Mar 2022 - Sep 2022
University of Virginia | Charlottesville, VA

- Consulted on scientific workflow automation, parallelization, and HPC performance; supported GPU-based machine learning for image processing and bioinformatics.
- Developed Bash tooling for software and container installation on the research cluster; implemented TensorFlow workflows using parallel GPU resources.

Optimization Intern, Rho AI Sep 2021 - Feb 2022
Charlottesville, VA

- Formulated supply-chain emissions reduction as a linear optimization problem using graph and matrix representations and a simplex-based solver.
- Built a hierarchical software model of supply chains to evaluate configurations and identify opportunities for emissions reduction.

Machine Learning Research Intern, STR May 2021 - Aug 2021
Analytics Division | Woburn, MA

- Developed a hierarchical Bayesian classification model for Monte Carlo simulation of discrete time series; improved mean correct classification rate by 15% in geospatial forecasting.
- Presented the research findings and applications in a companywide technical presentation.

TECHNICAL SKILLS

Programming: Python, Bash, MATLAB, R, SQL.

ML and numerical tools: PyTorch, TensorFlow, scikit-learn, NumPy, SciPy, pandas, Numba, CVXPY, Matplotlib.

Research methods: Diffusion models, variational autoencoders, latent generative models, score estimation, Bayesian inference, MCMC, optimal transport, inverse problems, uncertainty quantification.

Computing: Linux, Git, Docker, HPC clusters, Slurm, GPU computing, experiment automation.

SELECTED GRADUATE COURSEWORK

Deep Learning Theory; Statistical Learning Theory; Measure-Theoretic Probability; Numerical Linear Algebra; Optimization and Monte Carlo Methods; Kernel Methods; Partial Differential Equations; Functional Analysis.